

The Seneca Animation Model: Cross-disciplinary animation education program design for a changing industry

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ABSTRACT

Seneca Polytechnique, located in Toronto, Canada, first launched its three-year animation advanced diploma program in 2002 and regularly ranks as a global top-five program, producing or partnering on groundbreaking films like Chris Landreth's *Ryan* (2005, Oscar Winner), *Subconscious Password* (2013, Anney Grand Prix), and Aubrey Mintz's *Nothing to Say* (2018, multiple awards). Graduates have been in high demand in production studios around the world.

At the height of this success, the animation faculty proposed the unimaginable: scrap the program in its current form and start over.

Originally, the program was based on a familiar template that emphasized foundational drawing skills with increasing specialization leading to a capstone project in which students demonstrated skills aligned with production studios' needs. All students took two years of foundational instruction that focused on 2D animation production with classes in character animation, storyboarding, layout, character design, life drawing, acting and animation history. Toon Boom Harmony was introduced in second year, after a focus on paper flipping in year one. Students moving into the third year had the option choosing one of three specialization streams: 2D animation, 3D animation, and game art & animation.

While regular curriculum maintenance enabled graduates to be competitive in the labour market, specialization pathways became less clear because more methods were becoming cross-disciplinary between 2D, 3D, gaming, visual development, and visual effects. The goal of the new program was to create a wholly new cross-disciplinary animation education model in recognition of the changing conditions of the industry.

Our new program begins with a foundational year that develops student ability in animation art for any specialization. An example of this approach is a new first-year storyboard class, now named Film Language, which focuses on analytical assignments as opposed to creating storyboards for production. The related skills of boarding/previz are expanded upon in years two and three. We also increased digital workflows in our studios with specially designed animation desks that fit professional drawing tablets.

In the new program, students choose a specialization stream in year two, selecting one of four options: 2D animation, 3D animation, game art animation, and visual development. "Horizontal learning", in which students across all streams take common courses, promotes cross-disciplinary learning to enable practical awareness

of other production modes. For example, students across all four streams continue to mix for life drawing classes.

In year three, students continue in their specialization stream but will work on capstone projects that include team members across all streams. In this scenario, cross-disciplinary learning takes place in the collaboration of film projects rather than shared classes. Imagine a team of 2D students preparing their final-year film working with 3D students on some elements, game students to prepare real-time delivery of the final film in a game engine, and visual development students working concept art for the films of the 3D students. A "production weeklies" class in the final year enable the sharing of project concepts across all streams to stimulate self-organized collaboration.

To support these changes, Seneca invested approximately \$1.7 million to make significant enhancements in technology and facilities. This was required to accommodate many more students as the enrolment of the program will double from its original size. Upon full rollout of the new program in 2024, it will accommodate 450 students.

Besides solid production and artistic skills, the faculty pre-defined the goals for student and graduate professional expectations. These soft skills, including ability to work in teams, were foundational guides for the development of all new curriculum.

To support student success, they are enrolled in the following professional practices courses to strengthen soft skills in their first semester.

This process was almost entirely faculty-driven and supported by college administration in the investment of new facilities that mirrored contemporary production studios.

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Table 1: Competencies and outcome traits of students in new Seneca animation program

Competency	Outcome traits
Professional/ Interpersonal	Possesses a positive attitude and the ability to listen; Is a self-starter and effectively manages both time and expectations; Shows up on time, ready to work and possesses accountability for their actions; Always communicates consistently and professionally; Is creative, an enthusiastic collaborator and fearless in their chosen craft; Is respectful, courteous and a team player; Can clearly convey their own ideas.
Design & Colour	Demonstrates strong, effective choices for production art and design, based on a clear understanding of history, story, audience and themes.
Narrative	Demonstrates a thorough knowledge of storytelling as an art form, from themes, symbol, and metaphor to the use of more literal storytelling devices, character development and structure.
Movement	Demonstrates a clear understanding of animation principals and mechanics, as well as demonstrating good instincts through the execution of deliverables which demonstrate the required animation skills.
Composition	Demonstrates thoughtful consideration with regards to the principals of composition and design, ranging from the use and placement of shapes to direction of the eye as well as the use of structure, values, story and colour.
Life Drawing	Demonstrates capacity to connect life drawing with design, composition, animation, and story, and employs what they know about human, animal, and creature anatomy to their daily approach to design and animation.
Environment Design	Demonstrates consideration of the importance of environment and location, from the standpoint of design, composition and colour. They build a believable and engaging world for characters to exist and perform in.

Table 2: Description of first semester learning and professional foundational courses in new Seneca animation program.

ANIMATION BEST PRACTICES & WELL-BEING (7 weeks)
This subject introduces students to a number of animation survival skills aimed at improving good working habits, professional communication, time/expectation management and good overall mental and physical health strategies, enriching their overall experience and training.
GUEST ANIMATION PROFESSIONAL SERIES (7 weeks)
This subject introduces first year animation students to their chosen industry and to many jobs and professional elements that exist within in, in the form of guest speaker/Q&A series comprised of a variety of animation professionals from 2D, 3D, gaming and visual development. This subject aims to better educate our students about their chosen fields, exposing them to their new world sooner.